

Automating Organizational Research?

A Computational Framework for Process Theory Development

Abstract

Process theorizing has held a central place in organizational studies theory development by answering questions of *how*, *when*, and *why*. As organizational behavior becomes increasingly digitized and dispersed, organizational researchers have a novel opportunity to leverage data science for process theory development by combining micro, macro, and temporal views. This paper presents a guide for using data science techniques for process theorizing. Building on the grounded theory framework, we group computational techniques by their applicability at the different stages of theory development, from initial coding to verifying theoretical saturation. We argue that by mixing traditional qualitative sensemaking with computational techniques, grounded theory researchers can develop more holistic views, discern previously hidden constructs and connections, and prompt deeper inquiry, leading to more interesting and versatile theories.

Keywords: Process theory, computational methods, methodology, computational social science, mixed methods, grounded theory method, theory development

Introduction

In 2008, *Wired* Magazine published an article titled *The End of Theory: The Data Deluge Makes the Scientific Method Obsolete* (C. Anderson, 2008). Scholarly sources proclaimed the coming age of computational social science (Lazer et al., 2009). But after more than a decade, machine-learning models have not replaced the need for good theories (Lipton, 2016; Shmueli, 2010; Shmueli & Koppius, 2011). However, the jury is still out on exactly *how* to produce a theory from the ‘data deluge’, and the extent to which our methods as organizational researchers will change or remain the same (Mahmoodi et al., 2017; McFarland et al., 2016; Mills, 2018; Vaast & Urquhart, 2017).

Computational methods such as text analysis, social network analysis, and sequence analysis are beginning to be adopted in fields as diverse as psychology, medicine, education, engineering, and business (George et al., 2016; Kosinski et al., 2015; Lazer et al., 2009; Yu et al., 2011). Increasing access to fine-grained traces of social activity on a massive scale enables researchers to use new abilities to zoom in and out of socio-material routines, alternating between situated micro-level events and mechanisms, and emergent macro-level patterns (Gaskin et al., 2014; Inaba & Kakai, 2019; Venturini & Latour, 2010). Integrating and alternating between micro and macro perspectives this way helps researchers access both greater depth and breadth (Inaba & Kakai, 2019).

Despite the advances, traditional process theorists and organizational scholars have been slow to adopt computational tools for theory building, partly due to connotations of “positivism” and “variance theories” (Mohr, 1982; Shmueli & Koppius, 2011; Walsh et al., 2015a). Recent discussions such as “*Artificial Intelligence Meets IS Researchers: Can it Replace Us?*” (Loebbecke et al., 2019) highlight organizational researchers’ desire to

understand whether different steps of the research process are at risk of being automated, and what place remains for traditional research.

Rather than being alarmist, this paper argues that data science methods are most effective when interwoven with traditional process-centric and qualitative theorizing. To illustrate this complementarity in each step of the research process, we analyze data science methods through the lens of the constructivist grounded theory method (GTM) (Charmaz, 2006, 2014). For example, computational techniques can augment the task of developing codes via constant comparison (Glaser & Strauss, 1967) by rapidly comparing massive numbers of cases and concepts, even in highly complex, nuanced data. We find that by mixing traditional qualitative sensemaking with computational techniques, grounded theory researchers can develop more holistic views, discern previously hidden constructs and connections (Vaast & Urquhart, 2017) and prompt deeper inquiry, leading to more interesting and versatile theories (Agarwal & Dhar, 2014; Van de Ven & Poole, 2005; Walsh et al., 2015b).

This paper builds on a flurry of recent work developing the conceptual groundwork of computationally intensive grounded theory building (Berente et al., 2018; Inaba & Kakai, 2019; A. Lindberg, 2019; Vaast & Urquhart, 2017; Walsh, 2015; Walsh et al., 2015b). Prior work in this area has tended to focus on how computational techniques can be used during the inductive or descriptive steps of the grounded theory process. More recent work on diachronic analysis of digital trace by Pentland et al. (2020) has highlighted the large gap between researchers who *theorize* about process dynamics (Feldman & Pentland, 2003; Langley & Tsoukas, 2016; Tsoukas & Chia, 2002) and those who specialize in *measuring* processes and process change (Maaradji et al., 2017; W. M. Van der Aalst et al., 2003). We expand on these earlier works by showing where they fit into the larger framework of GTM and illustrating how nearly every step of the GTM process can be augmented by different types of computational techniques. In addition, we provide examples of specific techniques and algorithms for each step in the process, particularly focusing on those methods that exemplify the principles of process theorizing.

Process theorizing has held a central place in organizational studies theory development by answering questions of *how*, *when*, and *why* (Bacharach, 1989; Markus & Robey, 1988). Process theory perspectives have been especially influential in questions dealing with change over time such as technology adoption, organizational learning, competitive interactions, innovation, and strategic evolution (Feldman & Orlikowski, 2011; Langley, 1999). Rather than reifying static concepts and relationships, process theories get at the stories, dynamics, and processes that underpin organizational life, often from the point of view of those living it (Cloutier & Langley, 2020). Consequentially, process studies often employ qualitative methods to capture situated sensemaking and interactions in context (Feldman & Orlikowski, 2011; Langley & Tsoukas, 2010; Markus & Robey, 1988). In addition to building an inductive understanding of the phenomena, process theorists often contextualize their findings by drawing on a variety of sociological traditions including practice theory, structuration theory, actor-network theory, and so on (Feldman & Orlikowski, 2011).

Process theories have been defined in three main ways: process vs. variance models, narrative vs. logico-scientific paradigms, and process vs. substance ontologies (Cloutier & Langley, 2020; Langley & Tsoukas, 2010).

The first dichotomy proposed by organizational theorist Lawrence Mohr (1982) distinguishes between variance models (e.g. linear regressions) and sequential process models that describe what each event *could* lead to but avoids predicting what it *would* lead to. Within variance theories, the meanings of variables are thought to be constant over time, while their magnitude might change. By contrast, within process theories, the meanings of key concepts are subject to change as well (Ortiz de Guinea & Webster, 2017).

The second dichotomy contrasts narrative and logico-scientific paradigms. The logico-scientific paradigm uses concepts and generalizations to analyze the world “from a distance.” Narrative paradigms seek to capture the experience of living through a phenomenon in the moment by focusing on aspects of human emotion, meaning, and plot (Rantakari & Vaara, 2017).

The third dichotomy is the process vs. substance ontology. The process ontology, also called the “strong” process view, focuses on how *everything* (organizations, people, identities, etc.) is a process made up of continual micro-processes that reproduce certain patterns, as well as evolve and change over time (Langley & Tsoukas, 2010; Tsoukas & Chia, 2002). By contrast, the “substance” or “weak process” ontology conceptualizes entities, structures, or groups as discrete *things* that are capable of action or change, without unpacking their fine-grained continual day-to-day change and activity.

We will refer back to these distinctions when discussing the process-theoretic properties of different specific computational techniques, as each technique may be particularly suited to one view or another.

Based on the features outlined above, process theories exhibit certain strengths for organizational researchers. By providing context, stories, and action, they can grant readers a more on-the-ground experience of what organizational life is like. Stories can impart organizational culture and values, and can be used as a safe way for newcomers to be “shown the ropes” without getting thrown on the ropes (Pentland, 1999). By zooming out, researchers can draw the lines between the small actions or decisions that make up the daily life of the organization, and the high-level evolution of organizations through time. In this way, stories add meaning to the events, objects, and choices that make up peoples’ daily lives (Pentland, 1999; Weick, 1995). Process studies tend to highlight both the “deterministic” constraints and incentives that reproduce patterns over time, as well as the “free will” that humans have to respond in interesting, creative, and unexpected ways. Moreover, by highlighting *action* rather than *characteristics*, these studies provide readers the opportunity to imagine what they would do in similar circumstances, instead of merely predicting what would happen from a distance. By providing context through thick description (Geertz, 1973) rather than quantified tables, readers can judge for themselves how closely their own situation resembles the one presented in the study, and which patterns are likely to hold in both cases.

Despite its strengths, the hands-on work of developing process theories poses several practical challenges for researchers. The complexity, volume, and ambiguity of longitudinal qualitative data can make it difficult to know where to start or what themes to focus on, sometimes creating a sense of being suffocated by a tangled mass of information (Langley, 1999; Pettigrew, 1990). When cutting and condensing large unruly data into succinct narratives, researchers may worry they are overlooking important patterns or relationships (Walsh, 2015). In addition, while first-hand observation enables

interpreting events within their situated contexts, researchers may wish to buttress their observations with data beyond what they have personally observed (Dougherty, 2015; Markus & Robey, 1988). We believe these challenges can be addressed by combining traditional analysis of rich qualitative data with emerging data science techniques.

The paper begins by reviewing the constructivist GTM approach advocated by Charmaz (2006, 2014). Using Charmaz's framework, we analyze the recent literature on computationally intensive GTM. Next, we take a deep dive into each step of the GTM research process to show which categories of techniques complement each step of the theory-building process. In the following section, we discuss tradeoffs and challenges that are likely to arise for scholars using these techniques. We conclude with an illustrative example of how these techniques could have been integrated into a seminal study such as Barley (1986).

Relating Computational Approaches to GTM

Constructivist GTM

GTM was first laid out by Glaser and Strauss (1967) to systematize and legitimize inductive research in an era where hypothesis-testing dominated the social sciences (Bryant & Charmaz, 2007, p. 35). GTM encouraged researchers to look at the “raw data” of human experience and discover important new concepts, processes, and theories “from the ground up,” rather than relying on preconceived theoretical hypotheses and constructs. Over time, several branches of GTM have emerged, including Glaserian GTM (Glaser & Strauss, 1967), Straussian GTM (A. Strauss & Corbin, 1990), and constructivist GTM (Charmaz, 2006).

Compared to the objectivist-leaning Glaserian GTM and Straussian GTM, Charmaz's constructivist GTM emphasizes the role of researchers' subjectivity in the research process (Bryant & Charmaz, 2007, p. 131; Rieger, 2019). Rather than conceptualizing researchers as disinterested observers doing pattern-detection, the constructivist approach acknowledges that theory-building takes place within an ongoing theoretical conversation within a community of scholars (Bryant & Charmaz, 2010, p. 527). Constructivist GTM acknowledges that investigators and their audience both draw upon assumptions, interests, and vocabularies that influence how the “raw data” is interpreted, which emergent patterns receive attention, and which explanatory hypotheses are considered.

By mixing deductive and inductive analysis, constructivist GTM consciously makes use of *abductive* analysis (Bryant & Charmaz, 2010; Holton, 2011; Su, 2018; Tavory & Timmermans, 2014). Abductive analysis refers to “a type of reasoning that begins by examining data and after scrutiny of these data, entertains all possible explanations for the observed data, and then forms a hypothesis to confirm or disconfirm until the researcher arrives at the most plausible interpretation of the observed data” (Bryant & Charmaz, 2007, p. 186). Rather than starting with a blank slate, researchers following an abductive approach must be well-versed in the extant theories that cover their domain (Tavory & Timmermans, 2014; Van de Ven, 2007). That way, when entering the field, the researcher's attention will naturally be drawn to the unexpected, novel, or perplexing; indicating a blindspot or mismatch between prior theories and the empirical observations (Aron Lindberg, 2020). Noticing patterns that contradict or stand out against the fields'

expectations points theorists toward novel discoveries and “bold claims,” ensuring that the discoveries are a strong contribution to the field (Popper, 1963). This heavy reliance on the human researcher’s tacit knowledge (Polanyi, 1966, 2015) highlights one reason for the inherent impossibility of “fully automated” computational theory-building, as there would be no way for the computer to know which emergent patterns would be seen as surprising or perplexing, and which are already well-explained by prior literature.

In her description of GTM, Charmaz (2006) lays out several distinct steps of the GTM research process: data collection, coding, memo-writing, theoretical sampling, theoretical integration, and theoretical saturation. This is illustrated in **Figure 1**. Within the category of coding, Charmaz (2006) further distinguishes between initial coding, axial coding, and focused coding.

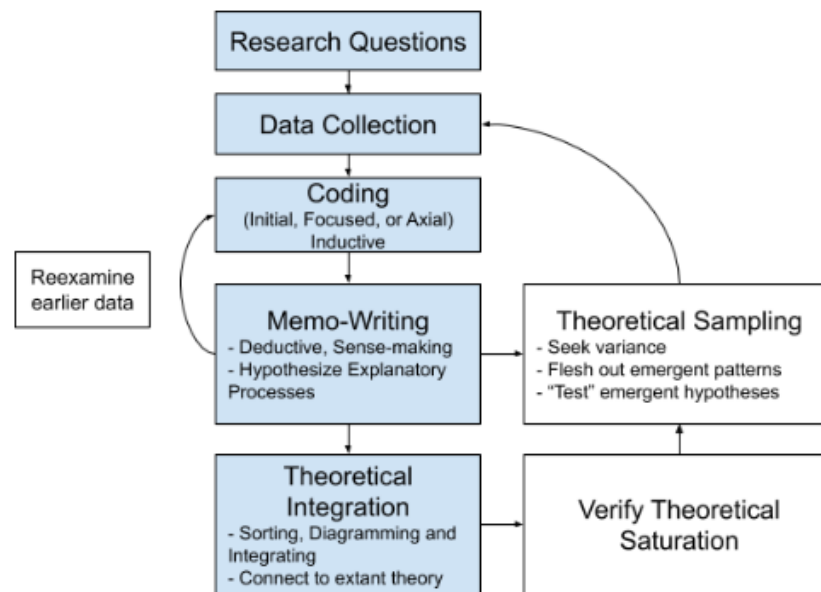


Figure 1. *The grounded theory research workflow.*

Initial coding, also called *open coding*, is the first-pass at “breaking data apart and delineating concepts to stand for blocks of raw data” (A. Strauss & Corbin, 1998). Traditionally, initial coding requires the researcher to slice data into “cases” or “instances” that can be analyzed line-by-line, categorized or labeled, and later compared and contrasted. As regularities among cases emerge, items with similar properties are grouped together into more general categories.

Axial coding refers to focusing on specific “axial” categories and fleshing out their properties, relationships, sub-categories, and associations (Charmaz, 2006; A. Strauss & Corbin, 1998, 1990; A. L. Strauss, 1987). By linking minor, situational codes to major concepts, axial coding reassembles data that was fractured during the initial coding to give coherence to the analysis. By consciously accounting for multiple situational factors, axial coding aids organizational researchers in gathering a complete view of their core processes. Some common relationships to identify include actions/interactions such as routines or strategic responses, the pre-conditions (who, what, where, when, why), and the range of consequences or outcomes (Charmaz, 2006, p. 61). This helps researchers

explain the internal consistency of each “ideal type” by identifying the underlying processes and causal dynamics, an important step in building middle-range theory (Doty & Glick, 1994; Walsh, 2015).

Once axial codes have been fleshed out and the core categories and processes are understood in detail, the researcher moves onto *theoretic coding*: integrating and finding high-level relationships between core categories. For process theories, macro-theorizing often takes the form of “visual mapping” to show the temporal and causal relationships between phases or practices (Langley, 1999). In our categorization of computational techniques, we blur the line between axial and theoretic coding. Although they deal with data at different levels of abstraction, from a conceptual point of view, they both are forms of “diachronic analyses,” drawing connections between pre-established codes to capture higher-order concepts or processes (Berente et al., 2018).

Focused coding is quite different from initial and axial/theoretic coding. Rather than generalizing lower-level data to discover new constructs or processes, focused coding involves scanning or reviewing prior data for instances of pre-established codes (Bryant & Charmaz, 2007). When done by hand, focused coding can both be a way of breaking rich data into parsimonious codes, as well as a way to refine the definition of the code itself as new variations or edge cases are brought into view and considered. Focused coding is often an intermediate step toward quantification. By categorizing rich data with conceptual placeholders, the researcher gains the ability to zoom out and consider relationships from different angles. Although focused coding typically refers to applying codes discovered during initial coding, it can also be used to tag data with codes from extant theories.

Memo writing is where much of the researcher’s sensemaking of the data takes place. Since data is often gathered longitudinally, investigators interweave analysis with data collection, writing memos to capture thoughts, intuitions, connections, emergent hypotheses, and potential directions to pursue (Charmaz, 2006, p. 73). Memos provide space for the researcher to hypothesize about the links between concepts and relate emerging theoretic categories to empirical examples. Writing memos during each analytic phase captures the associations while they are fresh, makes ones’ analysis progressively stronger, clearer, and more theoretical (Charmaz, 2006, p. 115).

Theoretical sampling is the practice of gathering more data as new questions, insights, and lines of inquiry emerge. Over time, different concepts or processes may seem to emerge as the “central story” of the study. Researchers must often gather more data to fully flesh out these constructs, learn what is going on in detail, and confirm or challenge their emergent hypotheses.

A construct is considered to be *theoretically saturated* when collecting additional data reproduces already-identified patterns and doesn’t add new information to the researcher’s understanding of the construct, process, or hypothesis. For purely qualitative studies, saturation occurs when additional data merely repeats old patterns and the researchers (and their editors) have a sense of fully understanding the core process. When incorporating quantitative methods, researchers may verify the validity of their understanding by formalizing it into a model and showing a strong quantitative fit with the data and that the model could generalize well using additional holdout data.

Theoretical integration involves tasks such as diagramming and sorting memos to visually illustrate the theoretical linkages between categories or processes uncovered in prior memos (Charmaz, 2006, p. 115). This may also be the proper time to explicitly draw connections to key concepts or processes described in extant literature, creating a coherence between the pieces of a study, and with the wider field as a whole.

Each of these steps is summarized in **Table 1**.

Table 1. *Major steps of the GTM workflow*

Research Step	Brief Description
Initial Coding	Developing new codes or categories by asking what data snippets “are a case of.”
Axial Coding & Theoretic Coding	Identifying middle-range constructs or processes. Identifying relationships between constructs, finding subcategories, or combining sub-elements into more macro constructs or processes.
Focused Coding	Mapping previously developed codes to large amounts of data.
Theoretical Sampling	Collecting additional data to flesh out emergent patterns, test emergent hypotheses, and explain anomalies.
Verifying Theoretical Saturation	Looking for anomalies that challenge the theory or model. Checking the robustness of emergent explanations and conclusions.
Theoretical Integration	Incorporating many parts of the theory together, and showing how it fits in with extant literature.

Can Grounded Theory Building be Quantitative?

In recent years, a growing number of scholars have adapted quantitative and computational techniques to work within the GTM framework (Aron Lindberg et al., 2016; Miranda et al., 2015; Vaast & Urquhart, 2017). A vigorous debate ensued in *Organizational Research Methods* over “What Grounded Theory Is”. The consensus found that GTM has been open to embracing both quantitative and qualitative methods since its inception in 1967, exemplified by the phrase “*all is data*” (Corley, 2015; Dougherty, 2015; Locke, 2015; Walsh et al., 2015a, 2015b). This consensus was noteworthy since historically, GTM has been adopted primarily by qualitative researchers (Charmaz 2006, p. xi; Walsh et al. 2015b). Several scholars went further, noting that although data mining and data visualization techniques could advance GTM scholarship, the methods have been slow to percolate into management journals (Dougherty, 2015; Walsh et al., 2015b). Even the most skeptical voices in the debate (Corley, 2015) allowed that quantification could be permissible within a GTM framework, as long as it accommodated a process-centric view that “engages the phenomenon from the perspective of those living it.”

Berente et al. (2018) published the first paper that formally outlines how to integrate GTM with novel machine pattern recognition methods. In it, they illustrate how computational techniques mirror inductive stages of grounded theorizing, such as (1) collecting observations, (2) defining higher-order constructs and taxonomies through “synchronic analyses” (e.g. clustering), (3) identifying patterns and relationships between novel and extant constructs from literature, and (4) producing “theory” by identifying causal or processual relationships among the constructs. Moreover, their paper highlights the constructivist nature of theory-building in general, by discussing how researchers’ vocabulary (i.e. lexicon) places observations into conceptual frameworks even before formal analysis or theory-building has begun. At the end of the day, any theory is just a linguistic compression of empirical observations, which is ineluctably embedded in our language of choice (Bacharach, 1989).

Our paper builds on this earlier work by expanding our scope beyond the inductive stages to include theoretical sampling and theoretical verification, as well as suggesting more process-theoretic techniques for the coding steps. The following sections elaborate on the use of computational methods within each of the stages of GTM.

Focused Coding

Focused coding refers to applying pre-specified codes or concepts to large amounts of data.

Although initial coding comes first in the GTM workflow, beginning the discussion with focused coding lays a foundation for understanding how computation fits into GTM.

Focused coding can be used both upstream and downstream from “initial coding” work. By tagging items with specific codes, it can aid the “constant comparison” needed for initial coding, especially computational initial coding techniques described in the following section. In addition, once rich data has been coded into set codes or variables using initial coding, researchers can use focused coded to extrapolate these codes to the entire dataset to discover relationships or visualize high-level patterns in the data.

There are three basic types of computational focused coding: *in vivo* coding, hard coding, and predictive coding.

Grounded theorists use the term *in vivo* codes to refer to the vocabulary used by participants to describe their world (Charmaz, 2006, pp. 55–57). *In vivo* codes often capture meaningful concepts within a field of practice, and have been traditionally used by researchers as a starting point for further analysis.

Researchers using computational methods or digitized data can draw on the *in vivo* codes present in their data in several ways. When dealing with natural language, researchers can record or code participants’ conversations using their own words. But *in vivo* codes can also be present in digital trace data, often in the form of meta-data. When using digital systems, programmers, users, or computer systems often “encode” bits of information that can be used as codes by researchers. Examples of this include, entering data into a database with pre-determined column names, tagging or categorizing documents on social media, or leaving trace data in logs describing what event took place.

As is the case with any grounded theory development, researchers should not import *in vivo* codes uncritically as theoretical constructs in their analysis. This is especially true when using digital archives or databases (Howison et al., 2011). Column names may be misleading, poorly defined, redefined over time, or simply auto-filled with inaccurate information during a data migration (Zahari et al., 2015).

While it is inadvisable to use *in vivo* codes as theoretical concepts, they can start out as low-level or middle-range codes to help summarize observations or provide context-specific evidence for a theory's findings (e.g., Barley 1986).

Applying *in vivo* codes to raw data can simply be a preliminary step of pre-processing the data for further analysis. When analyzing text documents, researchers apply “codes” that simply indicate the presence or absence of specific words in the text. This is often a crucial step for further computational text analysis (e.g., Lee et al. 2018).

Hard coding refers to computationally defining what a certain code looks like, and directing the computer to detect or measure the presence of the code in the digital data. Hard coding can be used to convert unstructured trace data into a format more amenable to analysis or to measure an aspect of the data that is of particular interest to the researcher. For example, Lindberg (2015) calculates the degree of variance of online routines based on the information present in the digital log data.

Common dictionary-based coding schemes use hard coded definitions to apply pre-determined concepts to text (Pennebaker et al., 2001; Tausczik & Pennebaker, 2010; Vaast & Urquhart, 2017). Althoff et al. (2014) use hard coding to operationalize concepts from prior literature such as “politeness,” “reciprocity,” and “status” in online communities using a mix of traces such as word-choice and up-votes.

As one might expect, hard coding can lean towards a positivist epistemology by extracting only the categories or variables of prior interest. On the other hand, the categories of interest can also emerge organically from close qualitative reading and analysis, and hard coding is one method to apply these codes to the rest of the dataset. For some reviewers or audiences, having an explicit operationalization of the coding scheme can also be a plus, lending the study additional credibility as well as adding a methodological contribution.

While *in vivo* coding uses labels already present in the text or log data, and hard coding calculates codes based on a set definition, predictive coding leverages machine learning to “train” a program to apply codes by showing it examples of previously coded data. Predictive coding can be useful for codes that have a “you know it when you see it” quality but would be difficult to pre-define computationally. In a recent example of predictive coding, Lu (2020) created a predictive classifier to determine whether individual Twitter accounts “transphobic rhetoric [...] under the guise of feminism” based on textual features and their positions in the network.

Predictive modeling generally assumes that the pre-labeled cases are roughly representative of the entire corpus. If the researcher is not sure if this is the case, they may opt to use semi-supervised learning. Semi-supervised learning techniques seek apply the same label to all items in a cluster, rather than simply finding the “most similar” pre-labeled cases (Van Engelen & Hoos, 2020). This may be particularly useful for labeling processes that evolved gradually over time. Compounding of differences may lead to quite

different features over time, while each instance of the process varies only slightly from prior performances. Predicative models that explicitly account for temporal “concept drift” may be quite useful as well (Žliobaitė et al., 2016).

Sometimes pre-trained models are made available to the general public, such as sentiment analysis models (e.g., Gilbert 2014), and other social-psychological measures contained in commercial packages such as *LIWC* and *SEANCE* that can pick up on features in the text such as anger, anticipation, disgust, fear, joy, sadness, surprise, and trust (Crossley et al., 2017; Pennebaker et al., 2001; Tausczik & Pennebaker, 2010).

Determining which features a predictive model should use is an important step for researchers to consider. For text data, there are many ways to convert the raw text into features and instances, such as using 200-word windows as instances of utterances, and constructing a feature for each word in the vocabulary (i.e. bag of words) (Aggarwal & Zhai, 2012). One downside of using bag of words models is that they can have limited interpretability. If the researchers want the models used for predictive-coding to be more interpretable, they can use the dimensionality-reduction techniques discussed in the subsequent section, such as latent feature analysis or clustering. If coding accuracy is more important than the interpretability of the features, the researcher may wish to use word embeddings, also discussed below.

Extrapolating hand-labeled codes to large-scale datasets can open up a variety of opportunities. Mapping codes to large corpora of texts such as online forums, researchers can gain a high-level view of what is going on over time and in different spaces. Zooming out this way to observe the overall picture can reveal interesting patterns or particular locations to “deep dive” into to understand qualitatively *how* the pattern looks at the ground level, and *why* it may be emerging in the first place.

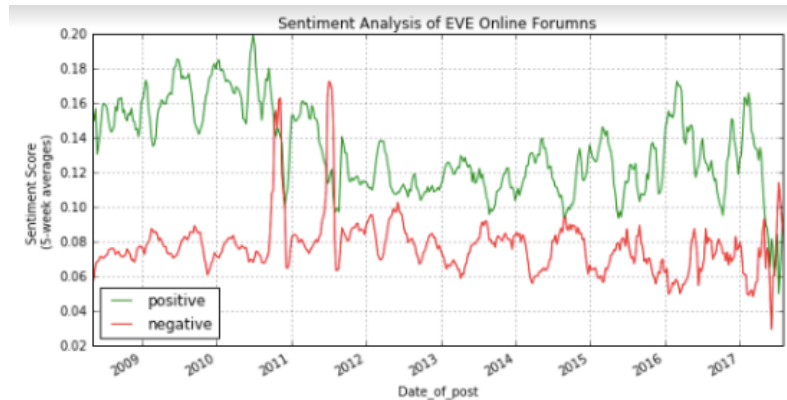


Figure 2. Sentiment analysis applied to a corpus of online discussion data. By applying labels to thousands of documents in less than a minute, one can easily see spikes of negative sentiment in late 2010 and 2011. This can help illustrate wider trends as well as guide where to concentrate for more focused analysis.

As an example, **Figure 2** shows a time-series of sentiment in an online community. The green line shows the level of overall happiness and the large red spikes indicate bursts of negative sentiment and anger. By identifying *when* and *where* this is taking place in the

community, we can zoom in to see what exactly is going on differently and what exactly is being said. Without a high-level view we may have missed out on some of these important moments of key concern to the community. Seeing the slow trends that take place over time is much easier this way as well, which might have been obscured otherwise if focusing only on periods of dramatic outburst.

Initial Coding and Axial Coding

Initial coding refers to the identification of low-level features and categories to distinguish cases from each other (Bryant & Charmaz, 2007). Axial coding typically involves elucidating the *relationships* between core (axial) categories and their associated elements, features, contexts, variants, mechanisms, and subcategories.

Although initial and axial coding are different steps in the research process, we address them together since there is a large overlap in the kinds of computational techniques that can be applied to both tasks.

Defining Cases and Features

Computational techniques automate the process of comparing cases to discover meaningful categories or dimensions, but first, the researcher must delineate what kind of cases they are investigating. Cases may consist of utterances of speech (segments of text data), sequences of actions (W. Van der Aalst et al., 2004), the trajectories of individual people, or entire communities (M. Zheng et al., 2017).

When coding cases manually, researchers “see through” the particularities of the data to identify what larger themes or processes are indicated by the surface-level information. By contrast, when using computational methods, the computer generally has no outside knowledge of the world or how it works, and merely groups items based on their similarity along the dimensions present in the data. Researchers must explicitly encode each item’s relevant properties or dimensions as codes or features (Berente et al., 2018).

Data can either be analyzed using the *in vivo* features present in the raw data (e.g. words or database entries), or features that are hard-coded or applied predictively as described in the previous section. The process of manually aggregating and transforming low-level data into higher-level codes is termed “feature engineering,” and can have a large impact on the later stages of analysis (A. Zheng & Casari, 2018).

Once the features and cases are defined and calculated, inductive analysis can begin. Induction involves identifying patterns of relationships among granular low-level data to create more abstract and generalizable categories, patterns, relationships, or processes.

Category Identification Techniques

The goal of category identification is to synthesize large amounts of data into a smaller set of meaningful conceptual categories for theory building. Dimensionality reduction can take many forms, from traditional clustering to more advanced methods such as topic modeling, network clustering, and latent factor analysis (Grimmer & Stewart, 2013). For many of these methods, we specifically discuss both temporal variants, and how the “static” versions can still be useful for temporal analysis.

How Computational Category Identification Aligns with Process Theory

Category identification and dimensionality reduction do not directly align with the “strong” process ontology or narrative paradigm that seek to tell stories from the participants’ point of view. However, many process studies still use the “weak” process ontology that identifies and reifies important concepts or categories from the observations. In this context, category identification techniques can fit comfortably into a process-theoretic framework in three main ways.

First, by doing a “distant reading” (Moretti, 2013) of the phenomenon, researchers can uncover the overall themes and trends that may have been difficult to untangle up close (e.g., Pentland et al., 2020). While techniques presented here may seem to emphasize “distal” over “proximal” points of view (Cooper & Law, 2015; Langley & Tsoukas, 2016), these methods can help call researchers’ attention toward potential blind spots of latent trends, and help direct collection of richer data on more novel, interesting, and generalizable patterns. In this way, inductive category identification can open the researcher’s eyes to new vistas and concepts that may have been previously closed to them.

Second, these techniques place events, situations, or characteristics into a relational framework, and aid ecological thinking by highlighting emergent patterns and their sub-elements (Tsoukas & Dooley, 2011).

Finally, while many of these techniques typically consider data to be static, running the analysis on sequential snapshots of data can reveal how the categories and processes evolve, illustrating the continual re-improvisation of social phenomena through subsequent performances (Feldman, 2000; Pentland et al., 2020).

Unsupervised Methods

During initial coding, the researcher is often trying to identify latent categories that most clearly capture and explain what is going on in the data. As the researcher often doesn’t know what categories they are looking for *a priori*, most of the computational methods we discuss here fall into the category of *unsupervised* methods. Unsupervised methods are essentially different methods for observing how patterns emerge organically from the data rather than having the computer extrapolate known categories to novel datasets.

Empirically derived groupings can be especially useful for discovering significant categories that may have been overlooked during deductive or qualitative analysis. For example, when using unsupervised analysis of political speech, Grimmer and King (2011) found clusters corresponding to four previously known types of political speech (“taking credit”, “advertising”, and “position taking”) but also discovered a new cluster of utterances that researchers interpreted as examples of “partisan taunting” that had been completely ignored by prior typologies.

Despite the hands-off nature of unsupervised methods, researchers still have ample flexibility to sharpen the focus of what kind of concepts are discovered. This is done primarily through defining what constitutes a “case,” selecting features, and choosing analysis methods. If researchers have a hunch that some aspect of the data is important, they can give those aspects additional weight or even filter out the parts data that do not correspond to the area of interest. As an example, when using census data to categorize

neighborhoods in terms of driving safety, it can be useful to include more features that have to do with commutes, such as commute length, number of cars per household, and how early people tend to leave for work, in addition to the other contextual features such as demographics, location, and geography.

In addition to defining cases and features, determining the method by which cases are grouped heavily influences the results. If the researcher wants to produce mutually exclusive groups, they can use clustering algorithms. If researchers think each case may be a mix of overlapping categories, they may employ topic modeling (Debortoli et al., 2016) or community detection techniques (Xie et al., 2013). Latent feature analysis can be used to identify dichotomous categories or dimensions. Discovering dichotomous categories can be especially interesting when doing text analysis to see what kind of themes end up as “opposites” of each other. Finally, there are several methods specifically designed to identify categories and dynamics of temporal sequences (Abbott, 1995; Aron Lindberg, 2015; Maaradji et al., 2017; Pentland et al., 2020; W. Van der Aalst et al., 2004).

Clustering

Clustering is typically used to discover mutually exclusive, categorical groups (Evans & Aceves, 2016; Grimmer & King, 2011).

Many common clustering algorithms such as *K-means clustering* group cases so that items from the same cluster are more similar than items from different clusters (Berkhin, 2006; Xu & Wunsch, 2005). This tends to create ball-shaped clusters, where some cases are more central or “typical” and others, more peripheral.

To understand what each cluster “means” qualitatively, researchers can both look at which features characterize the cluster centers, as well as zoom in qualitatively to look at the prototypical or peripheral cases in their native contexts. If clustering snippets of text, similarity may be indicated by using similar words, or other contextual information such as the timing, or who is speaking and who is being addressed. In this case, researchers can quickly make sense of what each cluster means by visualizing which words appear with higher frequency in each cluster. To get a deeper understanding, researchers can set up the program to print the text snippets in context, or link to the original documents for more in-depth qualitative analysis.

More specific varieties of clustering can be useful in different contexts. Hierarchical clustering methods can automatically create categories with subcategories (Johnson, 1967; Murtagh & Contreras, 2017). Other methods produce categories that overlap (Bezdek et al., 1984) or categories that indicate instances that are typical (i.e. central), atypical, or peripheral (Lingras & West, 2004; Mitra et al., 2006). Density-based clustering algorithms such as DBSCAN (Birant & Kut, 2007; Ester et al., 1996; Schubert et al., 2017) seek clusters of arbitrary shapes that are separated by low-density regions. An advantage of this density-based method is that it can handle noise effectively, it can detect “chains” of occurrences that gradually change over time or space, and the number of clusters to find does not need to be pre-specified by the researcher.

If scholars wish to create many clusters that are roughly equal in size rather than risk the results being dominated by a few giant clusters, we recommend using a quick method

such as Fast K-means (Sculley, 2010)¹ with a high number of clusters (e.g., 50) and pare the results down from there.

Clustering techniques can be particularly useful when doing multi-level analysis as is common in paradigms such as structuration theory (Giddens, 1986, 1991) and complex adaptive systems (DeSanctis & Poole, 1994; Miller & Page, 2007; Van de Ven & Poole, 2005).

Clustering “summarizes” low-level elements that can then be aggregated to characterize higher-level categories. For example, individual voters may be characterized by demographics or political preferences, but when moving up to the level of communities, the proportion of each voter type becomes a continuous feature characterizing the community. Similarly, individual snippets of text may be categorized into distinct clusters representing topics. At a higher level, documents can then be characterized as different kinds of mixtures of each topic.

From a more process-based view, clustering algorithms can group low-level trace data into *types of actions*. Sequences of those actions can be clustered into distinct *routines*, sets of routines may be clustered together into distinct *practices*, and sets of practices may be clustered into *fields*. By jumping only one level up each time, the researcher can maintain an interpretive handle on what is going on all the way down.

Many methods exist for quantitatively evaluating the clustering algorithm’s output (Jain, 2010; Jain & Dubes, 1988). However, we argue that it could be misleading for scholars to legitimize their findings by publishing the quality metrics of their cluster assignments (O’Neil, 2013). Rather, clustering should be used primarily as an input or inspiration for qualitative codes that are fine-tuned by the researcher to conform not only to the data at hand, but also to our knowledge of how the wider world works. It would be wise for researchers to keep in mind the famous quote, “all models are wrong, but some are useful” (Box, 1979). In this case, the clustering is only useful if it inspires the researcher and reveals novel trends that hold up under closer scrutiny.

Concept Discovery in Text using Topic Modeling

Whether working with field notes, interviews, or archival data researchers often find themselves dealing with textual data. As researchers comb through different documents to identify themes, they sometimes identify key words or phrases in the text and decide to plot the counts of these words through time or compare their relative usage by different segments of the organization.

A family of text analysis algorithms known as topic modeling presents an elegant computational solution that reduces the complexity of tracking words in text data while maintaining parsimony and context-awareness (Blei et al., 2003; Nikolenko et al., 2017). In addition to telling the researcher which words occur most prominently in each topic, topic-modeling algorithms also indicate what mix of topics are present in each document,

¹ Implemented at in Python at <https://scikit-learn.org/stable/modules/generated/sklearn.cluster.MiniBatchKMeans.html#sklearn.cluster.MiniBatchKMeans>

allowing researchers to quickly identify key topics and track their emergence and transformation through time.

The most well-known topic modeling algorithm is known as latent Dirichlet allocation (LDA) (Blei et al., 2003; Jelodar et al., 2019). Despite its popularity, there are several drawbacks with LDA that have limited its adoption by qualitative researchers: limited interpretability, confusion about how many topics to search for, inability to focus in on aspects of interest, and inherently erratic results even when running the same input through twice (Nikolenko et al., 2017). Despite the drawbacks, researchers have a few ways to adjust the parameters of the algorithm to get more meaningful outputs. Asking the algorithm to produce more topics can result in more interesting, fine-grained topics, with the common tradeoff of also producing meaningless “junk topics” that pick up on patterns in the noise. Setting it to produce fewer topics avoids the risk of finding patterns in noise, but may be too obvious or high-level to be interesting. With all its shortcomings, the topics and themes discovered by LDA may still be helpful sources of inspiration for potential codes as discussed above.

While LDA and its variants are the most well-known topic modeling methods, there are several alternatives. Topics can also be discovered by clustering *lexical networks* of co-occurring words (Rule et al., 2015). Discovering topics using lexical networks can provide more interpretability than the topic models like LDA that simply output lists of words. Within lexical networks, the position of words within the network can offer deeper insights into each topic, as well as revealing how different topics are related. Moreover, if a researcher wishes to unpack the qualitative meaning of a particular *link* or *cluster*, they can easily print and read the original documents or items that correspond to each link. In this way, the network structure grants qualitative researchers another easy means of “zooming in and out” of the data (Lazer et al., 2009).

Researchers can direct their investigation around themes of interest even more by using a variant of lexical networks that “seeds” the network around key themes. One way this can be done is by selecting a few key words *a priori*, and then constructing the network based only on text snippets that contain at least one of the key words. An example of this method is shown in **Figure 3**, which we constructed using a small corpus of process theory literature.

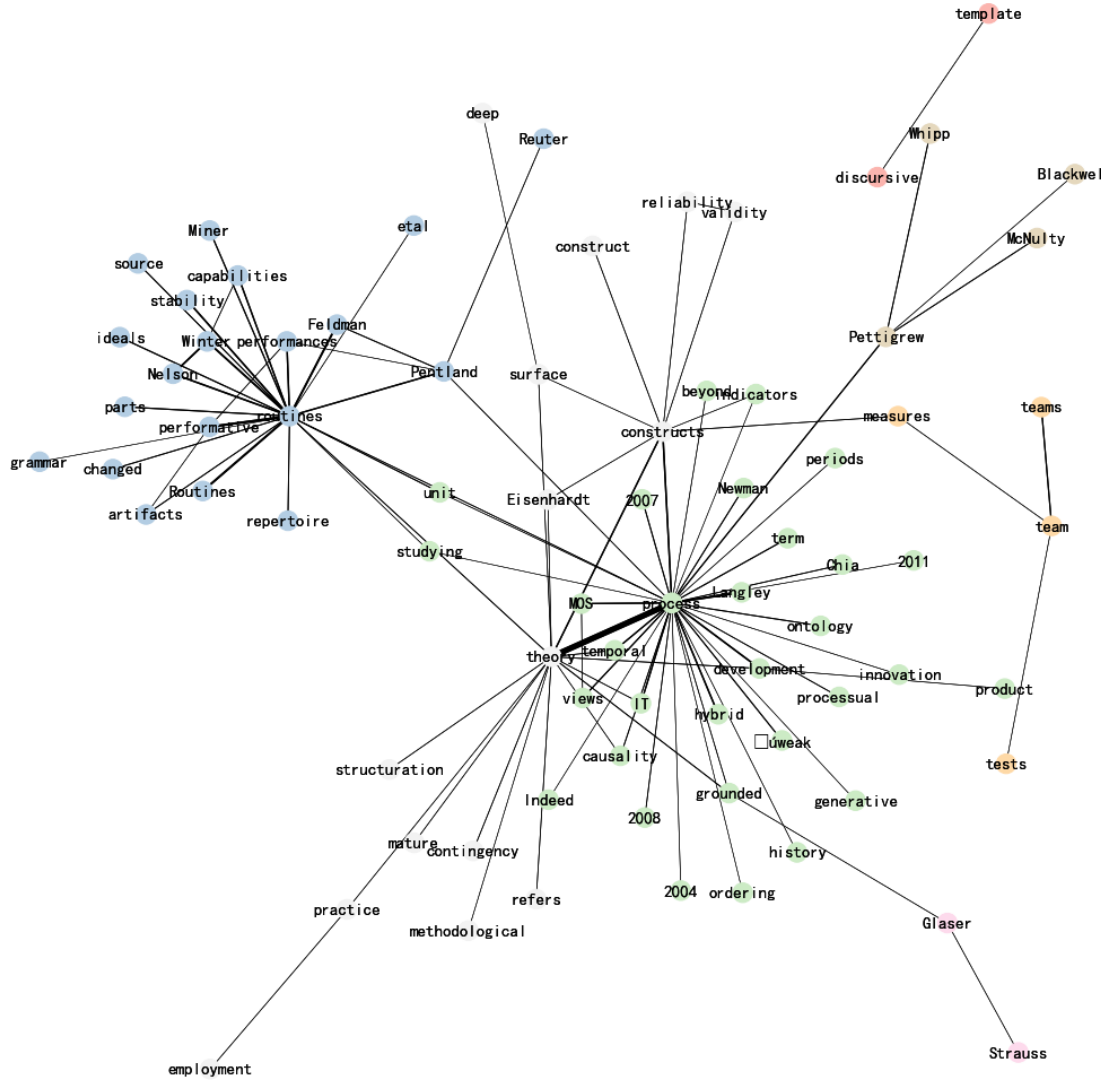


Figure 3. Lexical network diagram showing different co-occurrence between common terms in a corpus of process-theory literature. We seeded this network with the terms ‘process’, ‘theory’, ‘routines’, ‘constructs’, ‘Pentland’, ‘Pettigrew’, ‘practice’, ‘grounded’, and ‘measures’. Node color represents the results of spectral clustering, one way of segmenting a network into themes or categories.

In our example diagram, we illustrate how network clustering algorithms such as spectral clustering (Von Luxburg, 2007) can be applied to reveal higher-order topics. To control the complexity of the network visualization, we only display the top 150 links. Links are weighted by how often pairs of terms appear together in the text snippets. Term frequency, inverse document frequency (TF.IDF) was used to give higher weight to rarer words, which tend to be more meaningful than commonly used words (Aggarwal & Zhai, 2012).

Nikolenko et al. (2017) also have published a variant of LDA that makes use of “seeding” topics with different keywords and even assigning words to multiple topics, reflecting

potential polysemy. To our knowledge, there is no publicly available implementation of their method.

From a temporal perspective, looking at topics over time makes it possible to identify underlying themes or topical trends, and even identify particular documents that mark a shift towards new kinds of discourse (Nikolenko et al., 2017). From a “becoming” perspective (Tsoukas & Chia, 2002), it is not only possible to track the changes in the *prevalence* of particular topics from one time period to the next, but also to track the continual *evolution* and *emergence* of topics through time (Rule et al., 2015; C. Wang et al., 2012; X. Wang & McCallum, 2006).

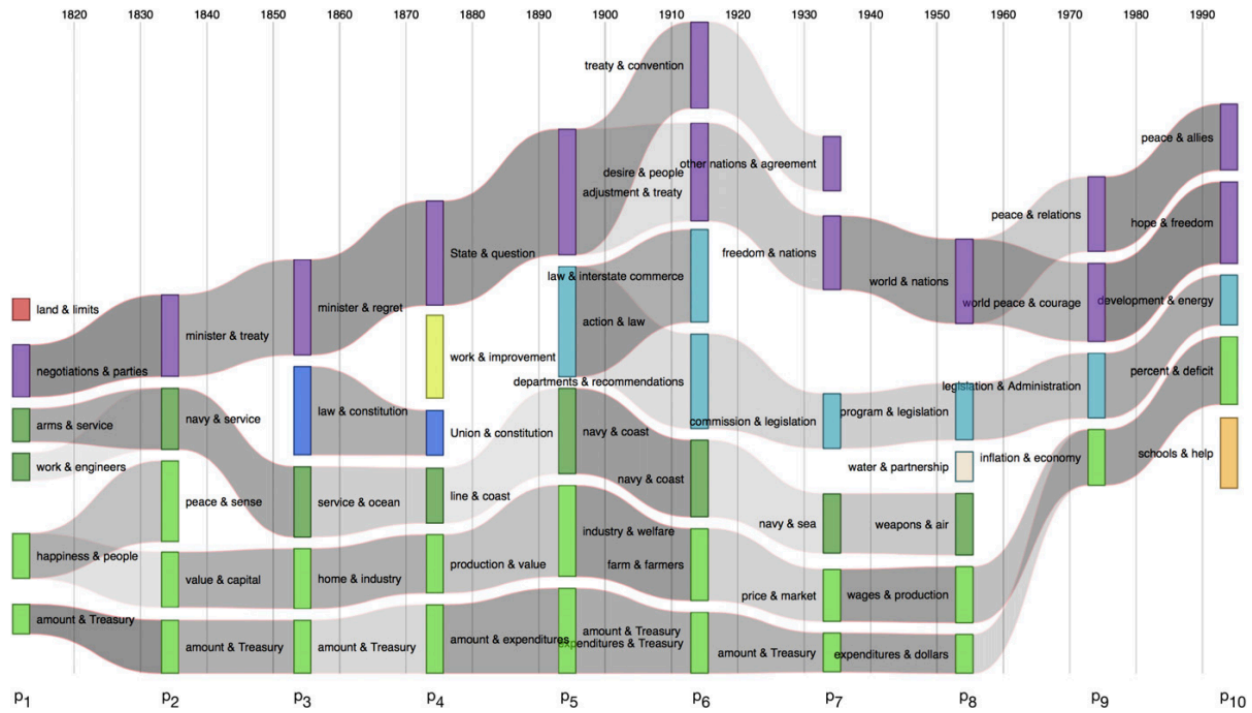


Figure 4 River chart of State of the Union addresses reproduced from Rule et al. (2015). Each box represents semantic clusters that emerged in each period’s State of the Union addresses. The grey flows between clusters in different time-steps indicate a high degree of shared vocabulary, with darker flows indicating greater similarity between the clusters. Streams that connect at any point in time are color-coded with the same color.

Two specific algorithms that build on LDA to account for the possibility of evolving topics over time include the topics-over-time (ToT) algorithm (X. Wang & McCallum, 2006), and continuous dynamic topic modeling (cDTM) algorithm (Blei & Lafferty, 2006). The main difference between these two models is that ToT focuses on sudden changes of topics’ prevalence while cDTM focuses on gradual changes within a topic’s composition over time. A downside of both of these methods is that they do not consider the possibility that topics may emerge, end, split, or merge at different points in time. Rule et al. (2015) address this issue by using the lexical network method described above, and showing how topics from subsequent time-periods can be connected into “lineages” by measuring the

overlap in each cluster’s “core words.” Once snapshots of clusters are connected through time this way, lineages can be traced that include emergence, branching, merging, and dissolution, as shown in **Figure 4**. By measuring the degree of change within a topic lineage from one time period to the next, researchers can also measure the rate of change or stability in the topics.

Discovering Latent Features in Data

Clustering divides cases into distinct categories. Topic modeling assumes that each case can be represented by a few themes. By contrast, latent feature analysis finds groups of *features* that seem to cohere. Latent factor analysis is neither new nor complex, and has been used by a variety of GTM researchers and settings.

Low-level features that co-occur can often indicate some abstract or underlying concept or generative mechanism that may not have been explicit or objectively measurable in the original data. The beauty of this method is that it not only reveals high-level features that may have been overlooked, but even if researchers were already aware of some of the concepts, it also reveals the constituent elements of each latent concept that the researcher may not have been aware of.

For example, the USA census captures hundreds of metrics about each zip code in the country, many of which strongly correlated. One latent category captures the “wealth” of the neighborhood, as indicated by correlations among income level, poverty levels, and occupations in management, science, and the arts. More surprising categories such as “urban periphery” are also revealed, and characterized by finding highly correlated metrics including leaving early for work, long commutes, recently moved there, more jobs in construction, warehousing, and transportation, and more foreign-born and Spanish speakers. While many readers may have been vaguely aware of the concept of an urban periphery, using these statistical methods can reveal aspects of life that may have not conformed to prior conceptions or expectations.

Illustrating to readers how such high-level concepts emerge from the specific low-level details can increase the theory’s versatility, trustworthiness, and generalizability (Osigweh, 1989). High-level concepts can even “future proof” the theory since the meaning of individual details can shift over time while the general conceptual framework holds.

One of the most commonly used dimensionality reduction techniques is principal component analysis (PCA) (J. A. Lee & Verleysen, 2007; H. Liu & Motoda, 2012; Wold et al., 1987). PCA is a well-known technique that finds groups of features that are highly correlated, while also ensuring that the discovered groupings are orthogonal. Each principal component is calculated by a weighted sum of the other features: $a_1 \times x_1 + a_2 \times x_2 + a_3 \times x_3 + \dots$ where x_i is the i^{th} feature and a_i is the weight of that feature. Researchers can interpret the meaning of each principal component by looking at which features have strongly positive or negative loadings. For researchers using the weights for interpretation this way, it can be useful to normalize each feature first (i.e. dividing each variable by its standard deviation) so that the weights can be meaningfully compared between variables that have different ranges (e.g. income could be denominated in thousands, while the number of cars per family would be in the single digits).

While similar to clusters, the conceptual categories generated by latent factor analysis need not be mutually exclusive: individual features could be included in several principal components, and unlike topic modeling, there is no assumption that individual cases are only represented by a small number of latent features. Another interesting aspect of latent dimensions is that they often produce dichotomies, with two meaningful concepts sitting at opposite poles of a spectrum, rather than a single concept being between “on or off” or “high or low,” as with clustering or topic-modeling.

When applying PCA to presidential candidates’ campaign speeches, we discovered opposing dimensions such as discussing “internal affairs vs. international affairs” and “focusing on policy and economics vs. focusing on the election itself”. Having distinct concepts at either end of the poles can be helpful and interesting; although it can also add unneeded complexity compared to clustering or topic-models.

As discussed above in the clustering section, the choice of data and features can strongly influence the kinds of latent features that are discovered. Researchers should be mindful about the initial choice of features to analyze if they wish to point the algorithm toward certain areas of interest and away from irrelevant information in the data that is not the focus of the study. While this kind of flexibility can be useful to investigators, it illustrates just one way the discovered features are not necessarily “objective.” Like any inductive analysis, the latent constructs are based on the choice of data and features that were fed into the analysis. Editors and readers should be wary of authors who defend their findings by saying “the computer spit it out, so it must be objective and real” (O’Neil, 2013).

Many of the aspects discussed here can be illustrated using Pierre Bourdieu’s exploration of cultural tastes and economic and cultural capital. Drawing on survey data Bourdieu (1998) used a form of latent factor analysis and data visualization called “correspondence analysis” (Le Roux & Rouanet, 2010) to map out the relationships between individuals’ occupations and tastes, as shown in **Figure 5**.

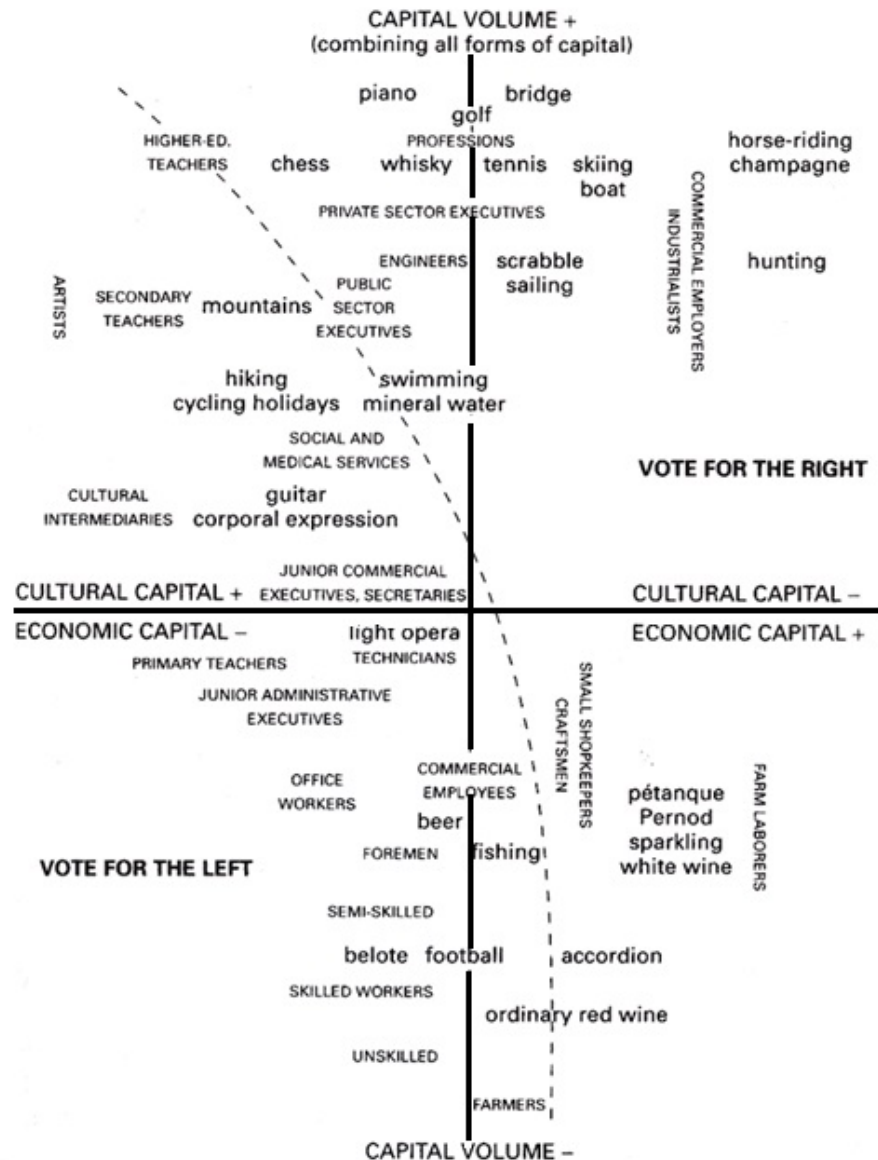


Figure 5. The space of social positions and the space of lifestyles (the dotted line indicates probable orientation toward the right or left). Reproduced from Bourdieu (1998). The UPPERCASE text refers to professions (the cases) and the lowercase text refers to different hobbies or tastes (the features).

The axes representing “relative cultural vs. economic capital” and “overall capital volume” emerged organically based on the correlations in the data rather than being hard-coded or pre-calculated before doing the analysis. By using correspondence analysis rather than PCA, Bourdieu was able to simultaneously display the “cases” (the professions) and the features (cultural tastes), as well as the resulting latent dimensions (overall capital volume and cultural vs. economic capital) that supported his theoretic claims.

Bourdieu's analysis illustrates how theoretically meaningful relationships and patterns can be grounded directly in an inductive analysis of the data, and how smart selection of the input data (e.g. tastes and professions) will produce latent features that are focused squarely on one's area of interest.

Bourdieu's analysis also shows how building up high-level features can "future proof" the theory, even though individual components may shift over time. In Bourdieu's analysis, we can see that consumption of "beer" was related to having low overall capital, but with the rise of the craft beer movement, we might expect to see this individual component rise in status over time, without challenging the overall framework or latent factors.

In addition to assisting sensemaking and generalizability, discovering latent features is often useful for subsequent tasks such as focused coding. Reducing high numbers of features into a smaller, more compact representation speeds up computational analyses and decreases the risk of over-fitting false patterns in the noise (Provost & Fawcett, 2013). This is especially true when using text as data, since representing each word as a unique feature means each document might be characterized by tens of thousands of features (the size of the vocabulary of the corpus). Rather than treating each word or phrase as unique and only counting documents as similar if they contain the some of the same words, dimensionality reduction can get at the general *gist* of each word. This way, documents with different words that convey the same theme would still be considered to be similar. We go into more depth on feature reduction for text in the following subsection on word embeddings.

To recap, using dimensionality reduction accomplishes three things simultaneously. First, it treats features (e.g. words or events) with a relational ontology by pragmatically encoding words that share similar meanings with similar latent features. Second, it may produce human-interpretable latent features and relationships. Third, it speeds up pattern detection on textual data or sequences with a high number of unique events.

Natural Language Processing with Word Embeddings

There are a few options for encoding words or phrases with latent features. Like PCA, latent semantic analysis (LSA) is an older method that looks at which words co-occur in a given document (e.g., a webpage, an academic article, an interview) to build a relational representation of each word (Evans & Aceves, 2016; Steyvers & Griffiths, 2007).

While PCA, LSA, and similar models use straightforward statistical methods to produce the composite features, recent advances in deep learning have led to a family of techniques known as word embeddings. Like latent features, embeddings create dense representations of text and words (Bengio et al., 2013; Goldberg, 2017). Unlike PCA, word embeddings are not created using simple weights on each word, and as such, the resulting dimensions are not interpretable on their own.

Word embedding algorithms seek to give similar coordinates to words that appear in similar contexts. The intuition is that two words that can be interchangeable within a given context are more likely to be synonyms or similar along other dimensions (e.g. part of speech, plural vs. singular, gender, temporal tense, etc.). While in many cases, words that are synonyms appear to have very close embeddings, there can also be cases where antonyms or other words are given similar embeddings because they tend to appear in

similar contexts. For example, “seasoned” and “unseasoned,” or “gay” and “lesbian” sometimes have similar coordinates despite not having the same meaning.

As stated above, the dimensions output by embeddings are not often meaningful or interpretable on their own. However, researchers can *define* meaningful dimensions along which the documents or words can be sorted by finding the coordinate vector between two words. For example, researchers can define a latent “gender” dimension by finding the vector that points between the coordinates of “he” and “she.” Bolukbasi et al. (2016) show how this method can be used to reveal stereotypes that exist in the way language is used, such as occupations that are typically associated with women or men. Researchers can also use embeddings to “reason by analogy” by adding or subtracting vectors from a word’s coordinates, as shown in **Figure 6**.

Extreme she	Extreme he	Gender stereotype she-he analogies		
1. homemaker	1. maestro	sewing-carpentry	registered nurse-physician	housewife-shopkeeper
2. nurse	2. skipper	nurse-surgeon	interior designer-architect	softball-baseball
3. receptionist	3. protege	blond-burly	feminism-conservatism	cosmetics-pharmaceuticals
4. librarian	4. philosopher	giggle-chuckle	vocalist-guitarist	petite-lanky
5. socialite	5. captain	sassy-snappy	diva-superstar	charming-affable
6. hairdresser	6. architect	volleyball-football	cupcakes-pizzas	lovely-brilliant
7. nanny	7. financier	Gender appropriate she-he analogies		
8. bookkeeper	8. warrior	queen-king	sister-brother	mother-father
9. stylist	9. broadcaster	waitress-waiter	ovarian cancer-prostate cancer	convent-monastery
10. housekeeper	10. magician			

Figure 6. Applying word embeddings to a corpus of text can reveal which words are more likely to appear in a feminine or masculine context in the text. In the *Extreme she* column, these occupations have the coordinates furthest in the feminine direction. On the right, embeddings were used to demonstrate analogies shifting “feminine” words to the “masculine” direction. Reproduced from (Bolukbasi et al., 2016).

Organizational researchers can either use pre-trained embeddings, or train embeddings that are more specific to their particular dataset. Building adequate word embeddings from scratch using your own data can be very data-intensive and computationally intensive. A viable alternative would be to download embeddings that have been pre-trained on external datasets such as Wikipedia and tweak the parameters using their particular corpus (Li & Yang, 2018; Qi et al., 2018; Yamada et al., 2018).

How can embeddings be used from a process theorizing perspective? Beyond simply improving performance on tasks like focused coding (Joulin et al., 2016), investigating how the embeddings themselves change over time can uncover changes in how language is used. Kulkarni et al. (2015) demonstrate several methods for modeling and visualizing the changing meaning of words over time and even offer a method to measure when statistically significant shifts have occurred, as shown in **Figure 7** with the evolution of the word “gay” from 1900 to 2005.

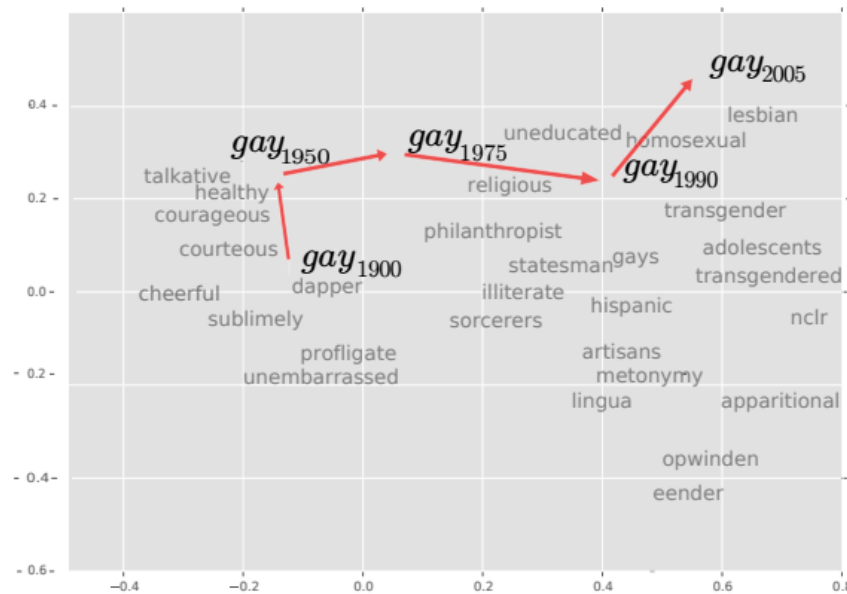


Figure 7. This graphic shows the words that were used in similar contexts to the word “gay” at different points in time. The authors produced the graphic by projecting each word’s embedding onto two dimensions of the latent semantic space. Reproduced from Kulkarni et al. (2015)

While word embeddings were originally meant to be used just to characterize individual words, it has been found that useable “document embeddings” can also be created by simply averaging the coordinates for each of the document’s words (Joulin et al., 2016). This implies that entire documents can be sorted or tracked along the dimensions derived from word embeddings, allowing researchers to track the shifts over time along custom-chosen dimensions. Detecting statistically significant shifts can also help identify pivotal moments or turning points within a corpus of documents, or help determine if the discourse from different parts of the community are beginning to converge or diverge.

Organizational Routines and Sequence Data

Identifying categories of organizational routines and temporal sequences is a special case of category discovery that is near and dear to organizational process theorists (Abbott, 1995; Pentland et al., 2020). Routines hold a special place in organizational process theory as they are seen as sources both of stability and change for organizations (M. D. Cohen & Bacdayan, 1994; Feldman, 2000). Routines and organizational processes are commonly used as units of analysis in organizational studies (Feldman & Orlikowski, 2011).

The arrival of the digital age means individuals’ actions are increasingly logged in minute detail, particularly in online spaces (Zuboff, 2019). This opens the door for researchers to study a wide availability of action sequences from different sources such as online discussion groups (Vaast & Levina, 2015), opensource collaborations (Aron Lindberg, 2015), and Wikipedia edits and revisions (Arazy et al., 2017; Geiger & Ribes, 2011; Heaberlin & DeDeo, 2016; Krishna et al., 2018). Because routines are repetitive and action-oriented, we generally have richer trace data of them than other latent or longer-

term or processes. Computational tools may prove especially useful for complex or hard-to-interpret sequence or log data, as manual or qualitative interpretation of action sequences can be more difficult than in-person observation. To make sense of emergent patterns, it may be helpful for scholars to engage in interviews with participants to get an insider view of what drives the patterned behavior.

Sequence clustering and routine mining has been a part of process theory discourse since at least the work of Abbott (1995). Over time, new methods have emerged that allow for increasingly complex processes to be discovered.

In a typical sequence clustering algorithm, the similarity between pairs of sequences is computed using their “edit distance,” representing the number of insertions, swaps, or deletions needed to convert one list to the other (Mitsa, 2010). Once the distance between every pair of sequences has been calculated, hierarchical or spectral clustering can be applied to group the sequences into as many or few groups as the researcher desires. This level of control allows both fine-grained or coarse-grained categories to be discovered (Wan, 2018). It should be noted that sequence clustering using edit distance is less useful when the researcher is trying to make strong causal claims such as establishing necessary conditions for subsequent steps (Mohr, 1982) since clustering by edit distance allows for some variation in the ordering. Other sequence clustering algorithms allow identification of exact matches of subsequences (Ferreira et al., 2007).

One downside of many sequence clustering methods is that they assume processes will be linear with no branching paths, cycles, or parallel, concurrent paths. While the linear sequence models discovered by sequence clustering are attractive in their simplicity, more complex process may be characterized by “action networks” with branching paths, cycles, or parallel tracks (Langley, 1999; Mintzberg et al., 1976; Nutt, 1984; Pentland et al., 2020; Pentland, Pentland, et al., 2017; Schroeder et al., 1989).

To adequately capture more complex processes, researchers may consider using Markov models or workflow mining algorithms. Markov chains calculate the conditional probability of switching between any two steps. Using these kinds of models, researchers can segment distinct patterns of behavior, even in routines with branches or cycles (Ferreira et al., 2007).

Like Markov models, “workflow mining” can reveal processes with branching paths and cycles (W. Van der Aalst et al., 2004). Workflow mining’s real strength is the additional ability to capture processes that contain parallel, or concurrent paths. Workflow mining algorithms analyze logs of time-stamped events to output action networks that capture the sufficient and necessary conditions to move from step to step. In its early iterations, workflow mining could only find a single “process flowchart” that could describe all the variations in a set of sequences and assumed that the processes themselves do not change over time. Recent extensions to this work do more to incorporate temporality by detecting points in time when the process was changed or replaced (Bose et al., 2014; Maaradji et al., 2017). Researchers should be careful when selecting which algorithm to use for this. Depending on which algorithm one chooses, some algorithms assume a punctuated equilibrium model with sudden transitions between processes rather than incremental changes, alternating changes back and forth between different processes, or gradual transitions in which the old and new processes coexist for a time.

Event-structure analysis (ESA) is a similar, more manual version of action networks that has been used by historians and sociologists to condense narratives into succinct networks (Griffin & Korstad, 1998; Heise, 1989). By combining the sequential actions in the empirical data with the researcher's knowledge and reasoning, ESA constructs networks of actions that show whether prior events are necessary (but not sufficient) condition for subsequent events to occur. ESA may be particularly useful for smaller data sets where more human reasoning is required to elucidate the causal structures (Cornwell, 2015; Mahoney, 2000; Ritschard & Studer, 2018).

How can workflow mining assist with process theorizing? First, workflow mining can break down the patchwork of sequences into distinct categories and routines. Categorizing action sequences into routines across the entire corpus of data allows high-level analysis to see where certain routines are more prevalent than others. The researcher might find that there are clusters of users that engage in one type of routine over another or see a drift in the kinds of routines that are being used over time or even diffusion of routines across groups of actors. Second, in addition to simply categorizing sequences, workflow mining's nature as action networks creates visual summaries and flowcharts of the actual action sequences, even if they branch, merge, or cycle. Analyzing the action networks is an emergent field of study among process theorists (Pentland, Pentland, et al., 2017; Pentland, Recker, et al., 2017). By investigating the network properties of the discovered routines, researchers can produce a richer understanding of how the "same" routine can be enacted in different ways. In addition, they may provide a more consistent way for scholars to measure and compare the routines they have studied which can provide the empirical basis for theorizing or conducting meta-analyses (Pentland et al., 2020).

While clustering can be useful for identifying significant groups, there can be challenges as well. In one sense, subjectivity is introduced simply in the act of choosing one particular clustering method (Fahad et al., 2014). It is highly likely that completely different categories would be produced if applying sequence clustering and workflow mining to the same dataset. Even within clustering alone, different algorithms would produce different groupings, especially on oddly shaped clusters, or clusters of differing density, and no single clustering method is guaranteed to be better than others (Jain, 2010).

Similarly, most clustering methods require the researcher to specify how many clusters are being searched for. The decision of how many groups to find is highly subjective, as quantitative heuristics often do not point clearly towards good cutoff points (Everitt, 1979; Ritschard & Studer, 2018; Uprichard & Byrne, 2012).

These issues show the relative strength of the qualitative scholar's expert knowledge. Rather than seeking validity and reproducibility in the "objective algorithm," scholars should use their personal judgment to decide which groupings are relevant to their study.

Axial Coding

Axial coding describes the step researchers take when they settle on specific concepts or processes as particularly important. To get a holistic picture of the process or concept, researchers look at its various antecedents, consequences, settings, and variants (or

subcategories). From this point of view, computational methods may have little to offer directly.

Another approach to axial coding could privilege inductive emergence rather than more researcher-driven approaches to constructing axial codes. In the spirit of exploring alternative methods, this section will describe two ways of using computational methods for axial coding. First, network analysis can be used to identify higher-order processes and categories, leveraging process theory's principal of relational ontology and ecological thinking (Langley & Tsoukas, 2016; Tsoukas & Dooley, 2011). Second, supervised learning approaches can assist in identifying subcategories or key elements of processes that have already been identified.

Axial Coding with Network Analysis

As explained above in the context of lexical networks, coherent clusters of words, properties, people, or events in a network can be a strong indication of a useful concept to investigate. Compared to category identification that seeks to group items based on *similarity*, network clustering can identify groups or processes based on how closely their elements *interact*. Finding tight clusters of relationships between items is likely highly indicative of persistent higher-order constructs or processes (White & Smyth, 2005).

The decision to focus on clusters of *relationships* between elements rather than feature similarity grants researchers flexibility to specify what kind of entities and relationships to consider in the first place. Some examples include interactions among individuals in social networks (Scott, 2017) such as interlocking directorates (Ma & DeDeo, 2018), citation networks (Agarwal, 2016; Ding, 2011; Shibata et al., 2008; Üsdiken & Pasadeos, 1995), and action networks (Abbott, 1995; Griffin & Korstad, 1998; Pentland, Pentland, et al., 2017; Pentland & Liu, 2017; W. Van der Aalst et al., 2004). The flexibility of network structures allows researchers to include humans as well as nonhuman actants (e.g. technologies, infrastructures, spatial elements, temporal elements, etc.) which may influence situations through their specific affordances, demands, agencies, properties, and requirements (Bryant & Charmaz, 2007, pp. 373–375; Latour, 2005).

Once the networks have been constructed, researchers can use network clustering or community detection algorithms to identify higher-order constructs from tightly coupled low-level elements. Depending on what the nodes and links represent, the constructs represented by a network cluster may have many different meanings: clusters in social networks can identify communities (Scott, 2017); tightly linked clusters of actions might represent routines of varying complexities (Pentland, Pentland, et al., 2017); clusters of words that co-occur in documents can represent different discursive topics or themes (Rule et al., 2015).

To center their analysis on particular axial codes, researchers can make use of the “network seeding” technique described in the preceding section and illustrated in **Figure 3**. Many clustering algorithms can divide up the network into different categories. See Xie et al. (2013) for an overview of different network clustering and community detection techniques. Agglomerative or hierarchical clustering may be particularly useful for identifying subcategories and super-categories within the network, as well as giving the researchers more flexibility to define categories at different levels of granularity (Fortunato, 2010; Shen et al., 2009).

If the researcher already has information about community membership, creating additional nodes representing each community can be helpful for tracing a network's temporal dynamics. Having nodes for both individual members and the groups they belong to creates bipartite networks, with “functional” nodes representing formal communities (e.g. online groups and meeting spaces), and “structural” nodes representing individual participants. Research by Zheng et al. (2017) has shown how using bipartite networks with individuals and groups can help make interesting inferences about the evolution of organizational dynamics. Their research described how as the growth of small, modular communities increased the overall resilience and redundancy of social networks, while increased consolidation of users around fewer larger groups resulted in information silos that reduced both system-wide coordination and resilience to external shocks (M. Zheng et al., 2017). We invite scholars to explore extending this kind of analysis beyond networks of participants. The concept of functional nodes could represent a wide variety of concepts that are shared between individuals such as regimes of justification (Boltanski & Thévenot, 2006; Dequech, 2008), shared practices (Wenger, 2011), or social identities (Hogg, 2016; Jenkins, 2014; Tajfel, 2010). Focusing on an axial category by making it a “functional unit” can be an interesting way to illustrate the temporal dynamics that the phenomenon undergoes over time.

Networks can be ideal tools for developing middle-range theories because of their ability to capture multiple levels of analysis simultaneously, from low-level nodes, intermediate-level clusters, and at the highest level, the properties of the network as a whole. Overall networks can be characterized through time by measures such as the clustering coefficient, average path length between nodes, and aggregates of node-level qualities (e.g. average number of links). Individual nodes can be characterized by their location within the network with measures of centrality, betweenness, or community-affiliation (Scott, 2017). Clusters or communities within the networks can be characterized by both network-level and aggregate node-level statistics (e.g. clustering coefficient, number of members, average centrality within the overall network). Plotting these metrics over time can reveal interesting temporal dynamics about how the phenomenon of interest is changing over time.

Network methods have been criticized by some process scholars for adopting an entity-oriented ontology rather than capturing the phenomenological feelings of experiencing reality in the moment (Langley & Tsoukas, 2016). On the other hand, many process scholars are comfortable treating processes and their constituent elements as units of analysis (Engeström, 2015; Latour, 2005; Nicolini & Monteiro, 2016; Ortnor, 1984, p. 148) or weaving in these abstracted theoretic depictions and diagrams alongside richer narrative accounts (Langley, 1999). We advocate this latter approach. Middle-range theories, like the kind that can be discovered with network analysis, are a crucial aspect of scholarship for their ability to identify general archetypes and processes, which provide support for building more powerful grand theories from the “ground up” (P. Anderson, 1999; Bryant & Charmaz, 2010; Merton & Merton, 1968; Van de Ven, 2007).

Axial Coding with Supervised Learning

The network methods described above represent one way of identifying how sub-elements that are strongly interrelated may indicate core processes or categories. Using interpretable classifiers can be another way to quickly identify antecedents,

consequences, context-dependence, or subcategories, especially when the axial category has already been identified.

Beyond simply extrapolating codes to large datasets, classifiers and predictive models can narrow down large sets of features to the ones that are most helpful for distinguishing one category from another (Puranam et al., 2018). Some specific techniques that can identify which features are most related to the target code include LASSO regression (Fonti & Belitser, 2017), random forests (Zhou et al., 2016), Shapley values (S. B. Cohen et al., 2005; Mokdad et al., 2015), association rules (Agrawal & Srikant, 1994), decision sets (Lakkaraju et al., 2017), and configurational analysis (Leonhardt et al., 2018; Ragin, 2009).

To illustrate how this could be useful, we can imagine researchers using feature-selection mechanisms with text data to generate dictionaries of words or phrases that are indicative of certain concepts or categories in the text (e.g., Hart et al. 2013). Publishing these kinds of lists becomes a methodological contribution and helps other researchers operationalize or measure the degree to which certain concepts are showing up in their own data. In addition to using these methods to find synchronic correlations between low-level features and high-level categories, the same methods can be used to trace diachronic relationships, highlighting which antecedents lead to which outcomes (e.g. Leonhardt et al. 2018).

Compared to predictive coding which can be incredibly flexible about the features it uses to label data, using supervised segmentation for axial coding requires that each case already be coded with relevant features (antecedents, consequence, participants, context, etc.). This data may already be freely available from the data source. At other times, these features may need to be applied by the researcher, possibly via focused coding. We don't recommend using this method for features that would be difficult or time-intensive to gather for every single case. If data on antecedents or consequences is slow to gather, it makes more sense to rely more on traditional qualitative methods.

Using Time Series to Identify Feedback Loops, Complexity, and Emergent Dynamics

Organizational process researchers often focus on unpacking the interaction between micro and macro forces shaping organizational life that may result in reproduction or transformation of familiar patterns of organizing (Pettigrew, 1997). Path dependence, emergent phenomena, and unpredictability of long-run outcomes are all common themes of process theories (Markus & Robey, 1988).

To help get a handle on organizational complexity, some process theorists have developed computational methods that predict the degree of centralization and coordination in a field by investigating the type of “noise” that arises in time-series of organizational behaviors (K. Dooley, 2002; K. J. Dooley & Van de Ven, 1999; Koput, 1997; Ott, 2002). Periods of highly random “white noise” coincided with decentralized, uncoordinated, or loosely coupled behavior. This may signal the absence of convergence around routines or equilibria, as actors experiment with many different actions. Periods of “pink noise” exhibit reversion to the mean and are associated with local level coordination, local constraints, and local feedback loops, for example, individual groups converging on set routines, and growing until they bump up against budgetary constraints. Finally, “chaotic” time-series behavior is associated with strong centralization or winner-take-all

dynamics, where a few key actors or decision-making strongly determine the dynamics of the entire system.

These techniques may not be suitable for every study, but they can help when deciding whether to focus on high-level decision-making, or low-level organizational behavior.

Theoretical Sampling, Saturation, and Verifying Conclusions

As the researcher's theoretical framework begins to emerge, it is important to search for edge cases, contexts, or sub-populations in the data that do not fall neatly into the proposed explanation. In the GTM, the practice of "theoretical sampling" describes the process of collecting and analyzing additional data about the phenomenon until all relevant variations of the core categories or processes have been observed and accounted for (e.g. different contexts, actors, subcategories, etc.). When subsequent observations do not add significant nuance or surprises, the study is considered to be "theoretically saturated."

This section will describe a few different techniques that can help guide and direct theoretical sampling efforts, especially when using large corpora of data. Traditional theoretical sampling often consisted of collecting additional data or field notes by hand. When using computational tools, it is not uncommon to collect more data than could be feasibly read (e.g. all of the conversations in a set of internet forums over a 10-year period). When practicing computationally intensive GTM, theoretical sampling could refer to either gathering new data, or zooming in to already-collected (but unread) data to analyze subsets of the existing dataset in greater depth and detail.

Looking for holes in the theory

The inductive techniques described in the sections above can help researchers identify strong patterns and core themes within a dataset. However, some of the most compelling evidence within qualitative research comes from investigating the outliers or "exceptions that prove the rule" (Tavory & Timmermans, 2014). Situations that break down or go awry are often deeply consequential for insiders and outsiders alike. Rather than being discarded as outliers that can be safely ignored, unusual situations may provoke participants to voice their implicit assumptions out loud, both for themselves, and for others in the situation (e.g., Garfinkel 1967).

The difference between a purely qualitative approach and a computational data-heavy approach is that you can "test" your framework against many more cases than you could realistically observe up close to find when these "exceptions to the rule" emerge. By operationalizing the constructs and mapping them to digitized corpora, the researcher can easily check how well the model applies to each group of cases and how well the chosen constructs help "explain what happens next." Adamopoulos (2013) used predictive models in this way to assess and evaluate the explanatory and predictive power of concepts identified via manual GTM.

While the goal of theory-development is not to produce a good predictive model (Gregor, 2002), the indication that the model performs poorly on a certain subpopulation may reveal hidden boundary conditions, mechanisms, or sources of variance that were not present or explicit in the majority of cases.

Along a similar line, converting the theory to a model and testing its error rate over time could reveal interesting turning points or moments of change. The most powerful way to reveal this would be to train a supervised model only on old cases and see how accurately it predicts future cases in each period. Even if the same constructs remain predictive, this test can reveal how their relationships change over time. Changes in the accuracy of the model likely mirror changes in what is happening on the ground. Looking at the error rate of the model could reveal sudden shifts, incremental changes, or even re-occurring situations.

Searching for additional cases

Beyond finding holes in the theory itself, predictive analytics can be useful early on to help gather relevant samples. Sometimes, scholars seek to find a representative or exhaustive set of documents such as during a systematic literature review. This is often done by selecting texts that fit a pre-determined set of keywords, and narrowing down by skimming papers' titles, abstracts, and conclusions (Brereton et al., 2007; Webster & Watson, 2002). In a sense, by doing this, the researcher is implicitly "coding" the texts ahead of time; assuming that the keywords represent the concept they have in mind, and excluding papers that may be related, but use different keywords to describe the phenomenon.

Predictive models can aid with searching or sampling documents with greater precision and breadth than key word searches. In this case, the researcher could train the model by providing a few examples of what she is looking for, and use a random selection of papers to indicate what she is not looking for. This way, such a supervised classifier could find a wider range of similar cases that more flexibly address the topic. If the model made "bad" suggestions, it would be simple to provide corrective feedback by retraining the model with hand-labeled data when it makes mistakes.

One downside of this method is that it assumes researchers have the "whole data set" at their fingertips. In reality, sampling often involves searching from disparate online sources or going back to collect qualitative data. In those cases, there are still a few ways to leverage data mining for data collection.

If using keyword search queries for documents in a particular category or concepts, feature selection tools such as LASSO may reveal better sets of keywords that are more indicative of the categories of interest (Fonti & Belitser, 2017). Alternatively, using techniques such as PCA, topic modeling, or lexical network analysis can reveal the "subcomponents" of a category that can be pursued in greater detail with either keyword search or qualitative interviews and observations.

Sharpening distinctions between emergent categories

In addition to manually investigating which items don't fit neatly into the emergent theoretical framework, another aspect of theoretical sampling is trying to sharpen the distinction between emerging conceptual categories. Using a subgenre of machine learning called "active learning" (Settles, 2012), researchers can automatically identify cases that seem to sit at the "boundary" between emergent categories.

Active learning tools can be particularly useful if trying to employ predictive focused coding, but more individual cases need to be hand-labeled as training data. Active learning algorithms can help select which instances would be *most useful* for labeling, either because the model has low-confidence about its predictions for these instances, or because there is a large subpopulation of “unlabeled” instances nearby. This also works on the assumption that the entire “population” of instances to be sampled from is accessible to the algorithm.

Finding the outliers that prove the rule

One downside of active learning or using a model’s error-rate to do theoretical sampling is that these tools would generally be used later in the research process when the researcher has already identified high-level categories to focus on.

If researchers want to identify edge cases or outliers earlier on, they can turn toward anomaly detection techniques (Chandola et al., 2009). Like clustering, anomaly detectors are “unsupervised” in the sense that the data does not need to be pre-labeled with high-level categories. Using the low-level features present in the trace data anomaly detectors specifically look for cases (or groups of cases) that stand out from the “typical” patterns.

Process theorists may be particularly interested in anomaly detectors that account for continual evolution of patterns and other temporal aspects (Chandola et al., 2010; Gupta et al., 2013). For example, a technique by Gupta et al. (2012) assumes that in any processual context, there will be many communities evolving simultaneously, so it detects event sequences that evolve in a divergent way compared to the other members of its cluster.

While “outlier detection” algorithms search for individual anomalous instances, “anomalous pattern detection” identifies groups of instances that seem to be generated via atypical underlying processes (McFowland et al., 2013; Neill et al., 2013). Qualitative process theorists may be particularly interested in text-based versions of anomalous pattern detection that identify “bursty” deviations from the smooth evolution of topics (Maurya et al., 2016).

Detecting points of significant change

Detecting significant events and turning points is can play a key role in the development of both process theories and traditional GTM (Bryant & Charmaz, 2007). “Significant events echo in memory, [... they] mark time and become turning points. A significant event stands out in memory because it has boundaries, intensity, and emotional force. Furthermore, a significant event captures, demarks, and intensifies feelings. [...] A significant event freezes and enlarges a moment in time. Because of inherent or potential meanings of self within the event, people grant obdurate qualities to it. They reify it. To them, the event supersedes past meanings and foretells future selves” (Charmaz, 1991). While many significant events may be identified via interviews with participants or spoken references to reified moments, unspoken events and shifts may still be present, and more easily detected via the use of quantitative time series methods.

Time series analysis has a long tradition in organizational process theory. Process theorists have used visualizations and statistical tests to identify and verify points of

significant transformation, focusing both on gradual change (Levina & Vaast, 2015; Meyer et al., 1995; Orlikowski, 1996) and abrupt change (Gersick, 1991; Oh et al., 2015b). Many qualitative process studies create “phase change” models, describing how events progress from one stage to the next. While in some cases the change-points are clear, at other times, leveraging time series analysis can identify or verify the moment of significant change within time series (Liu et al. 2013).

Oh et al. (2015) analyzes the dynamics of online protests by identifying shifts in Twitter hashtags such as “#Mubarak” or “#Egypt” during the 2011 Egyptian Revolution. The authors identified the moment of the hashtags going viral by identifying “structural breaks:” dramatic increases in the use of key hashtags over and above the model’s prediction. Oh et al.’s (2015) use of a Markov switching method allowed them to both verify that the changes they perceived qualitatively were indeed statistically significant, as well as to discover additional significant points of instability that they did not previously notice qualitatively. For example, they found that early on the growth of a certain hashtag (e.g., #Libya) escalated almost to the degree of creating a structural break, but did not quite reach that peak and de-escalated again. This observation led the researchers to zoom in and unpack it qualitatively, yielding interesting insights.

Oh et al.’s (2015) research on the Egyptian Revolution relied on looking at time series data one variable (hashtag) at a time and only considered transitions between two possible models of hashtag growth. However, several modern computational tools are available to process researchers that are capable of considering multiple variables simultaneously (e.g., Lavielle and Teyssiere 2006; Matteson and James 2014). Matteson and James (2014) propose a method that can identify significant changes in variables’ magnitude as well as their relationships (i.e. correlations) over time.

Finally, one simple way to detect statistically significant changes across many variables simultaneously is to use statistical process control charts (Bersimis et al., 2007; Santos-Fernandez & Santos-Fernandez, 2016). Statistical process control utilizes sets of rules that identify many types of shifts including gradual, abrupt, and cyclic. Compared to the Markov switching methods used by Oh et al. (2015), this approach may work with fewer time periods in a series, potentially making it amenable to data collected by hand about a given process, rather than from a time-stamped digital archive.

Time series can be generated either with archival data (e.g., time-series of financial information, frequencies of certain events, etc.) or with the codes and field notes generated by the researcher (e.g., Barley, 1986). When using researcher-generated data such as field notes, the researcher must be careful to record observations consistently, so that repetitive “old news” does not get ignored and left out of subsequent field notes.

Table 2. *Different computational techniques can be used to address different types of data or analysis. Temporal and process-centric techniques are **bolded**.*

Research Step	Brief Definition	Computational Methods
Initial Coding (Open Coding)	Developing new codes or categories by asking what data snippets “are a case of”	Clustering Topic Modeling

		Latent Feature Analysis Topics-Over-Time Sequence clustering Action Networks
Axial Coding	Identifying relationships between constructs, finding subcategories, or combining sub-elements into more macro constructs or processes	Network Analysis Network Evolution
Focused Coding	Mapping previously developed codes or categories to large amounts of data	Feature engineering Supervised learning Semi-supervised learning Supervised learning with concept drift
Theoretical Sampling and Saturation	Examining more data to flesh out emergent patterns and explain anomalies	Data visualization and exploration “Zooming into” unexplained patterns Supervised learning to sample relevant documents Identifying under-explained subsets Anomaly detection Accuracy over time Temporal anomaly detection Change-point detection

Challenges and Tradeoffs

The goal of this paper was to introduce a framework and suite of computational techniques for organizational process theorists. These tools support traditional qualitative analysis by unlocking the potential of using process-centric GTM with large archives of text or other digital traces. Nonetheless, we are well aware that there may be many hurdles for established process researchers seeking to take advantage of these techniques.

First, most of these techniques require a certain degree of fluency in programming. While many of the above techniques can be applied using easy-to-download libraries for R or Python, some may have to be implemented by hand. Engaging with a collaborator who is technically proficient but unaware of the assumptions and priorities of process theorists could create misunderstandings and conflicts that would have to be negotiated. On the

positive side, implementing these techniques from scratch enables researchers to make their own adjustments and methodological contributions!

Some of these computational techniques and methods may be unfamiliar even to more technically inclined researchers and reviewers. In particular, some of the temporal extensions, such as temporal topic modeling, are not commonly used even by data scientists or social scientists (Nikolenko et al., 2017). Mixed methods projects could be also challenging to publish, as there are fewer established evaluation criteria (Vaast & Urquhart, 2017).

Second, there are some problems inherent in using digital trace data. Data that the researchers did not generate themselves may have an imperfect “fit” to the real world (Agarwal & Dhar, 2014). When using log data from archival databases, column names may be poorly defined, or the way data was logged may have changed over time (Howison et al., 2011). Behavioral researchers need to consider the relevant ethical dilemmas related to privacy and consent when collecting and analyzing human social behaviors and interactions (Kosinski et al., 2015; Shmueli, 2017).

Third, many computational approaches would be most useful in large amounts of text or digital trace data. Such corpora may not be readily available to organizational researchers in a well-structured or time-stamped fashion. Analyzing the text of field notes may not be very useful to analyze computationally since they typically adapt to what the researcher already knows implicitly and are not a comprehensive view of the field. Since the qualitative researcher chooses where to focus the attention of his or hers interview or observation and what to record textually, using text analysis techniques on field notes may lead to self-reinforcing conclusions in data analysis rather than to new insights. Structured or semi-structured interviews might be useful to analyze for emergent themes, particularly if the samples were too large for humans to read repeatedly.

Finally, one salient limitation of the techniques presented above is an inability to “see inside the head” of participants to the extent that direct interviews or situated ethnographies might be able to do. One aspect of process theory involves investigating how the participants themselves conceptualize temporality (Schultz & Hernes, 2019), a task computational techniques would be relatively poor at performing. For this reason, we recommended pairing these techniques with more traditional qualitative methods and analysis, thus combining both breadth and depth.

Potential Value of Computational Techniques in the Context of Barley’s (1986) CT Scanner Seminal Study

We have listed many techniques here, but how might these fit in with a traditional ethnography of the type typically seen in organizational process research? To answer this question we look at Barley’s (1986) study *“Technology as an Occasion for Structuring: Evidence from Observations of CT Scanners and the Social Order of Radiology Departments.”* In this seminal paper Barley was already making use of some statistics to verify the theoretical story he was trying to tell. As the decisions made by radiologists and technicians emerged as a key element in his theory, Barley was able to keep track of this in more depth as he collected his field notes – allowing him to trace the proportion of decisions made by radiologists versus technicians over time (**Figure 9**). Barley also quantitatively counted the frequency of different discursive patterns (i.e. “scripts”) in his

field notes such as “unsought validation,” “role reversal,” and “accusatory questioning” (**Figure 10**).

Figure 2. Reproduced from (Barley, 1986). Proportion of operational decisions made by radiologists at two different hospitals (“Suburban” and “Urban”). Each data point represents the proportion of decisions made by radiologists vs. technicians on a given day. The vertical black lines represent external “shocks,” such as a new team of technicians being hired.

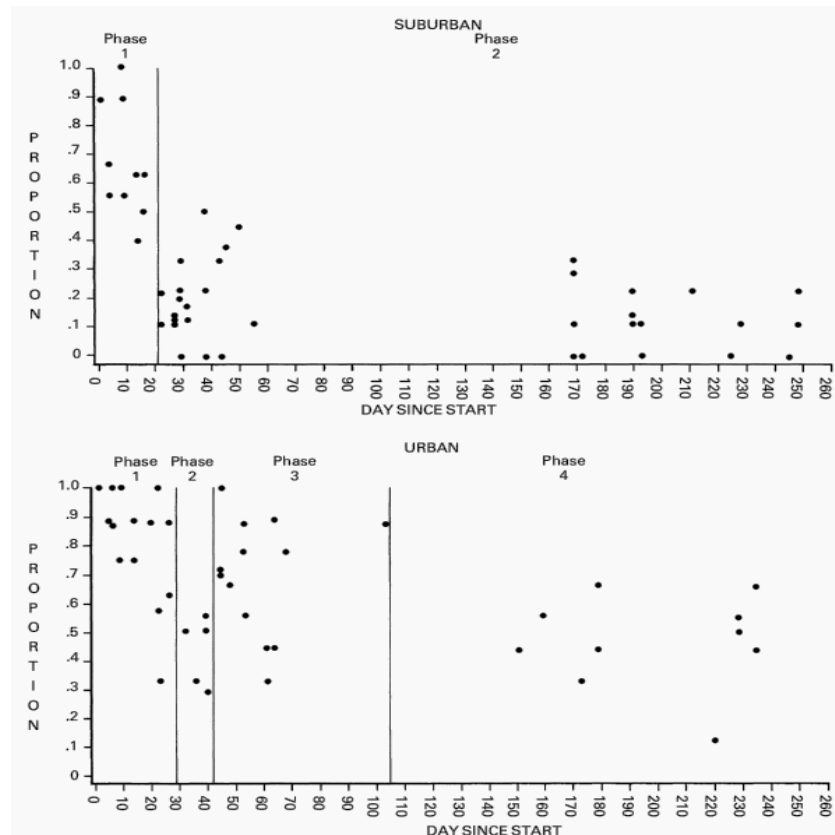


Figure 3. Frequency in field notes of scripts that gained ascendancy during phases of structuring at Urban and Suburban hospitals. Reproduced from (Barley, 1986).

HOSPITAL	PHASE	SUBURBAN							URBAN						
		Phase 1			Phase 2			DG	Phase 1		Phase 2		Phase 4		
		UV	AQ	PS	CT	RR	BT		CM	UC	DS	UCrit	AcQ	TC	ME
Suburban	1	6	9	0	0	0	0	10	0	0	5	0	0	0	0
	2	2	3	2	13	14	7	11	0	0	5	0	1	9	7
Urban	1	0	0	10	0	0	1	47	12	12	21	0	2	0	0
	2	0	0	1	0	0	2	14	1	7	13	11	6	0	0
	3	0	0	1	1	0	0	50	4	9	33	4	7	0	0
	4	0	0	0	0	1	0	13	0	3	11	1	0	11	10

Legend: UV = unsought validation
 AQ = anticipatory question
 PS = preference stating
 CT = clandestine teaching
 RR = role reversal
 BT = blaming the technologist
 DG = direction giving
 CM = countermand
 UC = usurping the controls
 DS = direction seeking
 UCrit = unexpected criticism
 AcQ = accusatory question
 TC = technical consultation
 ME = mutual execution

Simple quantification of codes allowed Barley to buttress his qualitative account of when the “phase transitions” occurred and easily illustrate the high-level trends for the reader. It also allowed Barley to include a simple hypothesis test (at the behest of his editors) to verify his assertion that the suburban hospital evolved through two distinct phases while the urban hospital evolved through four, based on the changes in who was making decisions day-to-day.

How might access to these advanced techniques have changed what kind of results Barley had gathered or analyzed?

First, even without additional digital traces, some text analysis techniques could be directly applied to the quotes in Barley’s field notes. For example, if it was hypothesized that there was an important emotional dimension in the interactions between technologists and doctors, the *SEANCE* text analysis package could measure the different emotions in each quote such as anger, anticipation, disgust, fear, joy, sadness, surprise, and trust (Crossley et al., 2017). With these additional dimensions, a more poignant set of constructs or turning points may have been discovered or suggested.

For identifying phase transitions, Barley originally based his analysis on memorable events and organizational members’ inter-subjective judgments as to when exogenous shifts occurred, such as a new team of technicians being hired. If any phase transitions did not depend on exogenous shocks, these may have been more subtle and easier to miss during qualitative analysis (e.g., Pentland et al., 2020).

To identify phase transitions, researchers could analyze changes in script frequencies using a Markov regime-switching methods (Oh, Eom, and Rao 2015), change-point detection in multivariate time-series (Lavielle & Teyssiere, 2006), or phase transitions between action networks (Pentland et al., 2020). Using the regime-switching model on the script frequencies would be a handy way of verifying that a change in staff resulted in a significant shift in the script usage, and also, would be able to pinpoint the amount of

time that it would take scripts to shift to the “new equilibrium” after the exogenous shock occurred.

In terms of coding, temporal topic-analysis techniques such as lexical networks could be leveraged to explore *how* key scripts or decisions evolved. In particular, if contextual “metadata” was included in the analysis (who said it, to whom, when, where, in what circumstance) then clusters of speech might reveal new scripts that Barley hadn’t originally noticed. These could act as an inspiration or a check to the original codes developed, potentially revealing an important new category or shift in the scripts’ language.

In addition to the data already gathered in Barley (1986), we can imagine what additional digital trace data could add to such an analysis.

Data on the evolution of social networks could be gathered either from the original field notes as well as interviews or digital traces provided by the organization such as networks of emails. Identifying steady or changing communities and social roles within the network could add a significant dimension of understanding how life the social world of participants was impacted by the shocks. Social network measures could be leveraged to identify how the changes had different impacts on individuals, groups, and the overall social network of the hospital. Adding digital traces might have expanded Barley’s view further into the rest of the hospital, potentially showing how changes in the radiology department were either ignored or complemented by changes in other departments or functions, such as patient handling.

Today, many companies are beginning to incorporate data-driven HR analytics initiatives to understand the social behaviors of employees, customers, and users (Burks et al., 2015; Gal et al., 2017). If the researcher was collaborating with the hospitals’ HR department, having data from employee reviews, hiring, quitting, changes in salary, etc. could illustrate how the findings impact key concerns for both the employees and the organization as a whole. Using anomaly detection, the research could both identify the high-level dynamics and simultaneously find the “exceptional cases” that can be followed up with to reveal the hidden dynamics and mechanisms that would otherwise go unstated and implicit (Tavory & Timmermans, 2014).

Conclusion

By identifying promising new computational techniques for detecting temporal patterns and zooming in and out of rich data sources, we offer a way of helping process theorists develop more insightful theories from rich digital data. Rather than ignoring a long history of qualitative scholarship, we describe an integrated approach where computational and qualitative analyses inform each other to develop a more authentic and more insightful picture of organizational complexity.

We specifically highlighted how computational techniques can be used for process theory building in ways that emphasize change, action, complexity, and multi-level emergent orders. By identifying critical turning points and anomalous behavior we provide tools to unpack the role of agency as opposed to fixed structured orders. Due to the inherent flattening that happens with computational analysis, we described how these tools can complement and augment rich qualitative data collection and theoretical sampling.

As these techniques gain traction in empirical studies, we expect new vistas for important theoretical development to open up, particularly in digital settings with thousands or distributed participants interacting asynchronously. By leveraging the GTM framework's constant iteration between emergent empirical patterns and theoretical insights, we are confident that organizational process scholars will continue to discover many new interesting and unanticipated questions and insights.

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